KIIT University — School of Computer Engineering

Smart Sales Analytics & Order-to-Cash (O2C) System

Project Report · Data Analytics Domain · SAP SD

Submitted By

Shreyas Das

Roll No: 23051786 Batch: 2023–27

Academic Year: 2023–2027

Subject: SAP Data Analytics Project Work

Abstract

This project presents a Smart Sales Analytics and Order-to-Cash (O2C) system built on Python-based data analytics technologies. It integrates the complete O2C business workflow — from customer inquiry to payment receipt — with an interactive dashboard offering real-time KPIs, trend analysis, and customer insights. Built using Pandas, Plotly, and Streamlit, the system shows how SAP SD processes and modern analytics can be unified into a single intelligent platform.

Table of Contents

Sec	Title	Pg
1–3	Introduction, Problem Statement & Objectives	2
4–5	O2C Workflow (7 Steps) & System Architecture	3
6–8	System Features, KPIs & Technology Stack	4
9–12	Data Schema, Data Flow, Enhancements & Conclusion	5

1. Introduction

The Order-to-Cash (O2C) process is the core revenue cycle of any commercial enterprise. It spans from customer inquiry to final payment receipt, covering order management, credit verification, fulfilment, invoicing, and collections — all central to SAP SD. This project merges the O2C workflow with a Data Analytics layer, creating an end-to-end system that executes the business process while providing real-time dashboards and KPI tracking through an interactive web interface.

2. Problem Statement

Most organisations rely on manual spreadsheets or costly ERP systems with no integrated analytics:

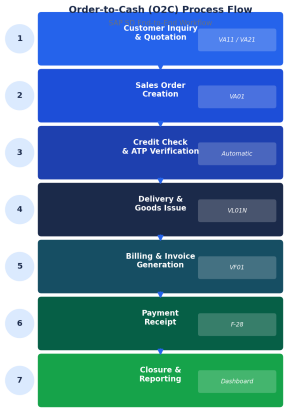
Issue	Business Impact
No real-time order visibility	Managers cannot act on delays until they escalate
Manual invoicing errors	Revenue leakage and customer disputes
Data silos across departments	Inconsistent sales, finance & operations data
No payment tracking	High Days Sales Outstanding (DSO) and cash-flow risk
Absence of dashboards	Decisions are reactive rather than data-driven

3. Project Objectives

- Implement the complete 7-step O2C workflow aligned with SAP SD best practices.
- Build an interactive sales analytics dashboard with real-time KPI monitoring.
- Enable CSV-based data ingestion for flexible, file-driven analytics.
- Deliver customer insights: lifetime value, purchase patterns, and churn risk.
- Provide product-level performance: top SKUs, margin analysis, and return rates.
- Create a user-friendly Streamlit web interface accessible to non-technical users.

4. Order-to-Cash Workflow — Step-by-Step Process

The O2C cycle runs as seven sequential stages, each mapped to a SAP SD transaction:



St	Process Name	SAP Txn	Description
1	Customer Inquiry & Quotation	VA11/VA21	Customer requests quote; system logs inquiry, products, and volume.
2	Sales Order Creation	VA01	Approved quote becomes Sales Order with pricing and delivery date.
3	Credit Check & ATP	Auto	Credit limit and Available-to-Promise check; order held or released.
4	Delivery & Goods Issue	VL01N	Goods picked, packed, shipped; inventory reduced via Goods Issue.
5	Billing & Invoice	VF01	Commercial invoice generated; tax, discount, and payment terms applied.
6	Payment Receipt	F-28	Customer pays; finance records payment against open invoice item.
7	Closure & Reporting	Dashboard	Cleared items archived; analytics updated with revenue and KPIs.

Fig 1: O2C Process Flowchart (left) with SAP transaction codes (right)

5. System Architecture

The system follows a five-layer architecture ensuring modularity, scalability, and clear separation of concerns:

Layer	Technology	Responsibility
Data Input	CSV / Excel	Raw sales, order & customer data ingestion
Processing	Python + Pandas	Cleaning, transformation, aggregation
Visualisation	Plotly	Bar, line, pie, scatter, funnel charts
Workflow	Python Functions	O2C step execution and status tracking
Presentation	Streamlit	Dashboard UI, filters, multi-page navigation

Fig 2: Layered System Architecture

6. System Features & Dashboard Modules

Sales Overview

- Total Revenue & Orders (MTD/YTD)
- Revenue by region & category
- Daily/weekly trend charts
- Inquiry-to-Order conversion rate

Customer Analytics

- Customer Lifetime Value (CLV)
- Top 10 customers by revenue
- Repeat purchase & churn risk
- Segment-wise spend analysis

Product Performance

- Best-selling SKUs by volume & value
- Slow-moving inventory alerts
- Gross margin per product
- Return rate tracking

O2C Process Monitor

- Order-to-Ship cycle time
- Days Sales Outstanding (DSO)
- Invoice accuracy rate
- Payment collection status

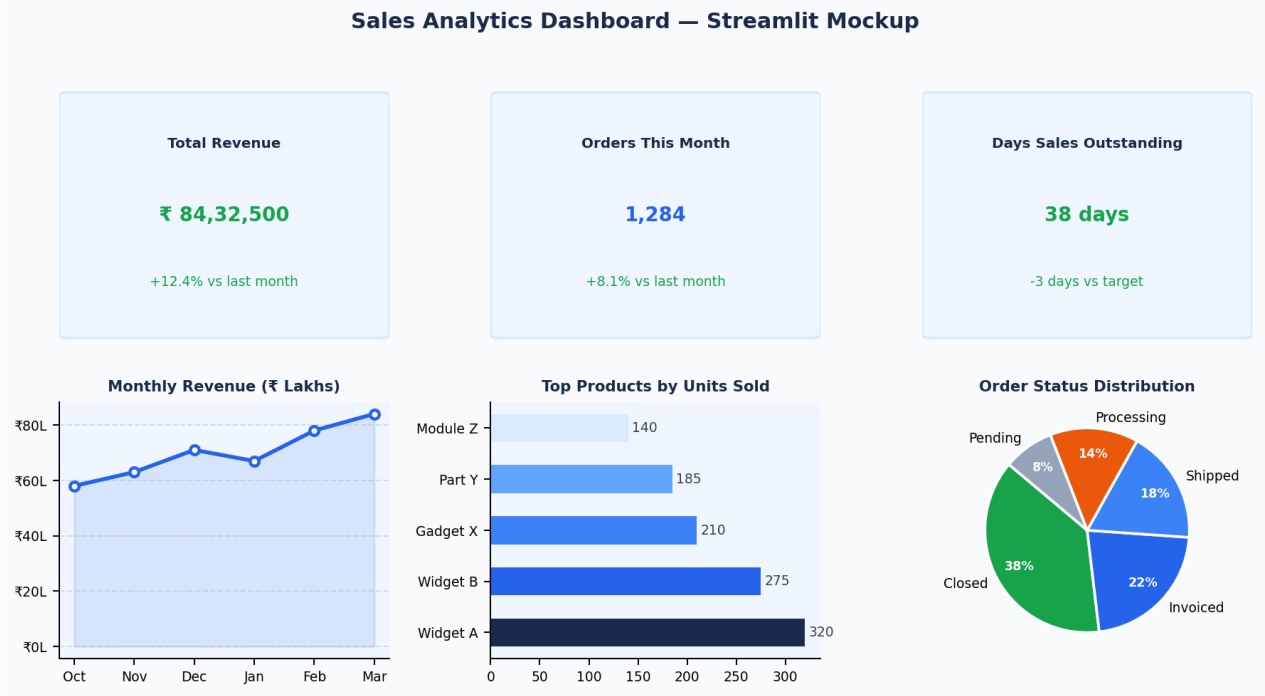


Fig 3: Streamlit Dashboard — KPI Cards, Revenue Trend, Top Products & Order Status

7. Key Performance Indicators (KPIs)

KPI	Definition	Formula
Total Revenue	Sum of all billed sales	SUM(Invoice Amount)
Days Sales Outstanding	Avg. days to collect payment	(AR Balance ÷ Revenue) × Days
Order Fulfillment Rate	% orders delivered on time	(On-time ÷ Total Orders) × 100
Gross Profit Margin	Profitability per sale	((Revenue – COGS) ÷ Revenue) × 100
Invoice Accuracy Rate	% of error-free invoices	(Correct ÷ Total Invoices) × 100

8. Technology Stack

Tool	Ver.	Role in Project
Python	3.11+	Core language — all logic, processing and integration
Pandas	2.x	Data ingestion, cleaning, transformation, aggregation
Plotly	5.x	Interactive charts: bar, line, pie, scatter, funnel
Streamlit	1.x	Web dashboard framework — UI, routing, file uploads
NumPy	1.x	Numerical operations and array-level computations
GitHub	—	Version control, collaboration, deployment pipeline

9. Sample Data Schema (CSV Input Format)

The system accepts CSV files with the following column schema as its primary data source:

Column	Type	Description	Example
order_id	String	Unique order identifier	ORD-00142
customer_id	String	Customer account number	CUST-5021
order_date	Date	Date the order was placed	2024-03-15
product_sku	String	Product stock-keeping unit	SKU-BLU-XL
quantity	Integer	Units ordered	50
unit_price	Float	Price per unit (INR)	1,250.00
total_amount	Float	Quantity × Unit Price	62,500.00
order_status	String	Current O2C stage	Invoiced
payment_status	String	Payment received flag	Cleared
region	String	Sales region	East India

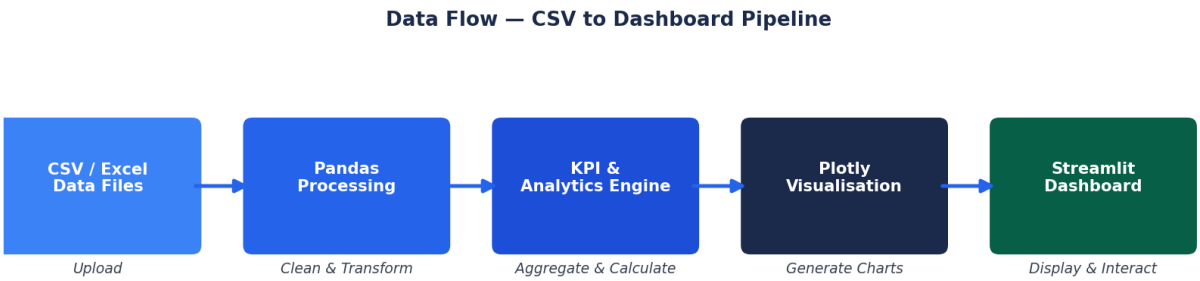


Fig 4: Data Flow — CSV → Pandas → Analytics Engine → Plotly → Streamlit

10. Future Enhancements

Enhancement	Description
SQL / PostgreSQL	Replace CSV with a relational DB for persistent multi-user storage.
AI Demand Forecasting	Integrate Prophet/scikit-learn for sales prediction & trends.
Live SAP API	Connect via SAP OData REST APIs for real-time data streams.
User Auth & RBAC	Role-based access for managers, analysts, and admins.
Automated Alerts	Email/SMS alerts when KPIs breach defined thresholds.

11. Conclusion

This project demonstrates how the O2C process — a cornerstone of SAP SD — can be realised through a Python analytics system. Unifying the 7-step workflow with an interactive dashboard bridges ERP logic and modern analytics. KPIs such as DSO, fulfilment rate, and CLV surface actionable insights, while Streamlit ensures accessibility. The project fully satisfies KIIT SAP Data Analytics requirements for an end-to-end O2C scenario with step-by-step documentation.

12. References

[1] SAP SE. (2024). SAP SD O2C Process Docs. help.sap.com
[2] McKinney, W. (2022). Python for Data Analysis. O'Reilly.
[3] Plotly Technologies. (2024). plotly.com/python
[4] Streamlit Inc. (2024). docs.streamlit.io
[5] KIIT University. (2024). SAP Project Guidance Document.
[6] Pandas Team. (2024). pandas.pydata.org